25CSPCC303IAE-II

0 3 OCT 2026

Seat No. : .....

Student Name : .....

St. John College of Engineering and Management, Palghar,

Maharashtra

Autonomous Institute

Internal Assessment Examinations, October 2026

B.Tech. Computer Engineering - SEMESTER III /IT/AIML/DS

25CSPCC303 - Microprocessor & Computer Architecture

St. John College of Engineering & Management SJCEM PALGHAR

Time : 1 Hour **Maximum Marks : 20**

Instructions

- All Questions are compulsory and carry equal weightage of 5 marks
- Assume suitable data wherever required

Q1	Attempt any 5 out of 6 Questions	Maximum marks(5)	Course Outcome	RBT Level
	a)What is the function of Port C?	1	CO4	L2
	b)Name any two flags in the EFLAGS register.	1	CO5	L1
	c)State the purpose of a TLB in a processor.	1	CO6	L1
	d)Why are RAM and ROM assigned different address ranges in a microprocessor system?	1	CO4	L2
	e)Explain the role of the Memory Management Unit (MMU) in the architecture of the 80386.	1	CO5	L2
	f)What is the main objective of Hyper-Threading Technology?	1	CO6	L2
Q2	a)Apply Single Transfer Mode for transferring data from an I/O device to memory.	5	CO4	L3
	OR			
	b)A printer is connected to IR2 and generates an interrupt. Apply the 8259 interrupt sequence.	5	CO4	L3
Q3	a)Explain the functions of the Instruction Fetch and Decode Unit in Pentium.	5	CO5	L2
	OR			
	b)Explain the Virtual 8086 Mode of the 80386.	5	CO5	L2
	-lyze the evolution of Intel processors from m 4 and explain the major	5	CO6	L4

	architectural improvements introduced in each generation.			
	OR			
	b)Analyze how the architecture of Pentium 4 supports faster instruction execution.	5	CO6	L4

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